INSPIRECON 2026

CAPSTONE PROJECT

ST. ANDREW'S COMMUNITY HOSPITAL

SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

BACKGROUND

Who We Designed For

Nurse Manager:

Spend >3 hours creating each roster, balancing requests, staffing and policies

Ward Staff:

Want fair and transparent way to request and view their shifts

Current Challenges

- Manual Scheduling in Google Sheets
- Double Data Entry & Compliance Risks
- Lack of Transparency & Slow Approvals
- Limited Off Site Access
- No Automated Alerts to Staffing Issues

Why It Matters

- Alleviate Manager Overload & Admin Burden
- Alleviate Staff Frustration & Low Morale
- Prevent Understaffing & Policy Breaches

INTRODUCING DUBY

Your shift buddy



Final Roster Output

1) Deployment Readiness:

Validated, compliant schedules ready for immediate use by hospital staff

Aggregated Information

1) Real-time Overview:

Real-time overview of staffing distribution across roles and shifts

2) Operational Balancing:

Identification of understaffed or overstaffed shifts for efficient resource reallocation

Data & Constraints Setup

1) Unified Datasets:

Integration of profiles, leave, and shift requests into a centralised optimisation engine

2) Rule Mathematicalization:

Translates MOH guidelines and operational rules into mathematical constraints

3) Dynamic Adaptation:

Adapts staffing requirements dynamically for holidays and surge periods based on manager input

Smart Interface

1) Workforce Optimization:

Weekly/bi-weekly overview designed specifically for managerial decision-making

2) Visual Intuition:

Colour-coded shifts for quick identification and decision-making

3) Flexible Editing:

Easily adjust and reassign shifts with minimal effort

Request Overview

1) Centralized Submission:

Submit leave and shift preferences directly through the centralised request interface

2) Transparent Communication:

Instant visibility of approvals and rejections through an integrated notification system

MILP Solver Model

1) Constraint Satisfaction:

Generates optimal schedules satisfying all hard constraints

2) Equitable Efficiency:

Balances fairness and operational efficiency

CP-SAT

1) Feasibility Insurance:

Fallback roster generation for scenarios where constraints cannot be fully satisfied

2) Scheduling Robustness

ACHIEVEMENTS & COMPARISONS

	TIME TAKEN	COMPLIANCE	FAIRNESS	EFFORT
BEFORE	4 hours	Manual, Error-Prone	Subjective	High (manual rework)
DUBY	✓ **15 mins**	✓ **100% compliant**	✓ **Data-Driven**	✓ **Low (automated)**

[DUTY BUDDY] BY

Wong Zhi Hung Xavier
Lee Peck Yeok
Tay Pei Yun
Mikhail Yasoputra Jeffrey
Tania Koh Tze Ern
Lee Wei Jie
Do Viet Anh

TRAILBLAZING A BETTER WORLD BY DESIGN.